

TRACK · DIGITAL ASSET MANAGEMENT

PRANA प्राण

PROCESS RESPONSE & ADAPTIVE NETWORK
AGENT

INDUSTRIAL FLEXIBILITY INTELLIGENCE

How much electrical flexibility can an industrial plant safely afford to use?

Turning process physics into safe, economic electricity
decisions with the production objective untouched.

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IIT Mandi

Platinum Jubilee Innovation Hackathon
MC²⁺ · Oil India Limited · IIT Kharagpur

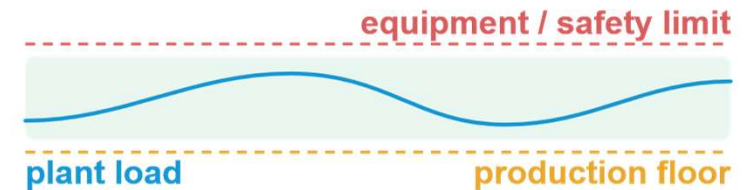
THE MARKET MOVES

Power price within one delivery day



THE PLANT DOES NOT

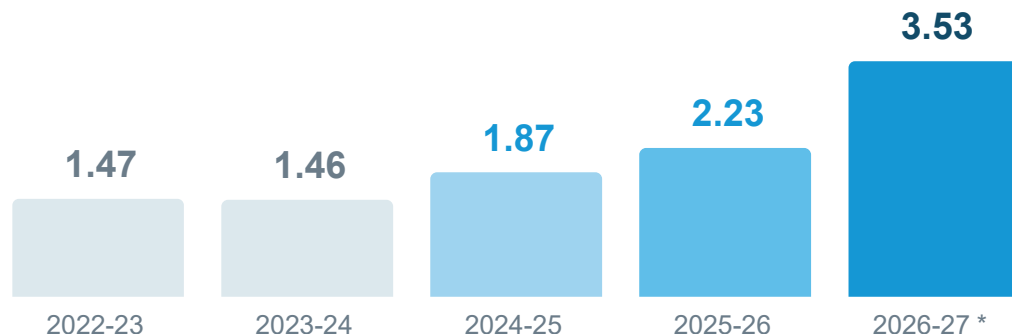
Load, inside the safe operating envelope



Electricity is no longer a flat-cost input

IEX real-time market, Maharashtra W2 mean evening price divided by mean solar-hour price, on every settled 15-minute block since April 2022.

EVENING-TO-MIDDAY PRICE RATIO, BY FINANCIAL YEAR



21.1%

of the year now clears below ₹2/kWh, up from 4.1%

₹10/kWh

the regulatory cap, hit repeatedly in the evening peak

152,542

settled blocks behind these figures nothing modelled

Price volatility creates an economic opportunity — but only if plant flexibility can be quantified safely.

* April–August 2026 only India's peak-demand and peak-solar months. Seasonally biased upward; it indicates acceleration, it does not measure it.

A plant cannot simply follow the electricity price

THE MARKET ASKS FOR

Low price

→ consume more

High price

→ consume less

Two rules, both of them ignorant of what the plant is making.

THE PLANT ANSWERS TO

Safety

Inventory

Equipment limits

Production

Ramp rate

Quality

Downstream demand

Electricity cost ranks below every one of these and always will.

A price signal alone is not an operating instruction. The same ₹/kWh can produce completely different decisions depending on what the plant is making, where it is operating, and which constraint is binding. PRANA converts that process context into an actionable energy decision.

THE FLEXIBILITY GAP

Electricity markets know the price. Plants know their process. The missing layer is the decision between them.

PRANA is that layer. It quantifies how much electrical flexibility a process can safely and economically use, and returns one of three answers each morning:

SHIFT

HOLD

DO
NOTHING

DO NOTHING is a valid, frequent and important result.

PRANA discovers flexibility it does not assume it

Potential flexibility sources plant-specific validation required. None of these is assumed flexible until its own physics says so.

Refinery

Compressors, utilities, intermediate tankage as a process buffer.

CANDIDATE

Fertilizer

Utilities, electrolysis where present, ammonia and urea buffers.

CANDIDATE

Chemicals

Batch scheduling freedom and intermediate inventory between units.

CANDIDATE

Hydrogen

Electrolyser operating envelope and downstream storage pressure.

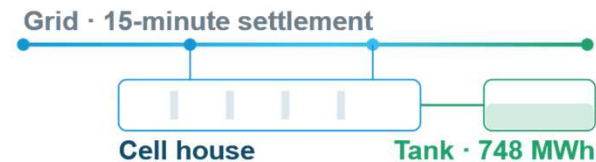
CANDIDATE

Chlor-alkali

DEMONSTRATED · 120 DAYS

Rectifier load, caustic inventory, chlorine offtake as the binding downstream constraint.

Where the storage already sits



Four rectifier trains, one caustic tank. The tank is the state of charge: run the cells hard on cheap power, then coast through the evening on stored caustic. Tonnes shipped stay the same.

The physics changes from industry to industry. The flexibility-intelligence framework remains the same.

From process physics to flexibility

PRANA uses a plant-specific physics model. Chlor-alkali is our demonstrated implementation — the equations below are its, not everyone's.

DERIVED, NOT FITTED

$$V = V_0 + k \cdot i \quad \text{cell voltage law}$$

$$q \propto i \quad \text{Faraday's law}$$

$$P(q) = \alpha q + \beta q^2$$

Convex by electrochemistry, not by parameter choice. Splitting output into a turn-down leg and a rebuild leg therefore always burns more kWh than steady operation that excess is the true physical cost of flexibility.

The published cell curve implies β of 0.30–0.45; this model uses **0.21**, so we charge ourselves the larger round-trip loss.

Process physics

Convex load curve, energy per tonne, power factor



Operating envelope

Hard interlocks and equipment limits as bounds



Production & downstream

Tonnes committed, offtake that cannot be stored



Inventory & ramp limits

Buffer capacity, rate of change the plant tolerates



Flexibility cost ϕ

₹/MWh to move load, output held constant

NPC India, Chlor-Alkali Sector Manual, p.28 (voltage law) and p.29 (empirical cell curve, $E = 2.41 + 0.329 \cdot i + 0.24 \cdot \log i$). Linearised for the MILP by 16 tangent hyperplanes; measured error 0.025%.

PRANA gives the plant a number it didn't have

$\phi(\Delta MW, \Delta t)$ The cost to this process of shifting electrical load by ΔMW for Δt hours, while respecting every production and engineering constraint.

LOW COST

Usable flexibility

The process can absorb the movement for less than the market pays for it. Recommend **SHIFT**.

HIGH COST

Probably uneconomic

Membrane wear, diverted product and reactive-power penalties eat the spread. Recommend **HOLD**.

INFINITE · UNAVAILABLE

Do not touch

The flexibility does not exist at that depth or duration. ϕ returns infinity rather than a number. **DO NOTHING**.

WHERE THE VIRTUAL BATTERY FITS

Quantify safe flexibility



Represent as power × duration

The virtual battery is a **representation of measured flexibility** never the purpose of the plant. We do not assume a process is a battery; we find what it can spare, then describe it in grid units.

PRANA does not ask “how much load can we shed?” It asks “how much flexibility can we safely afford?”

Data to setpoint, in five stages that already run

01 · DATA

Settled market panel

152,542 IEX blocks, 1,588 complete days. MERC Case 75 of 2025, post-remand, read from the primary order.



02 · TWIN

Cell-house physics

Convex power curve, caustic tank as state of charge, chlorine balance, H_2 -in- Cl_2 as a hard bound. Emits ϕ .



03 · FORECAST

Quantile price fan

LightGBM q10/q50/q90, conformally widened out of sample. MAE 1,072 against three baselines.



04 · DECISION

Stochastic MILP

96 blocks against landed cost, $CVaR_{95}$ risk term, demand charge inside the objective. CBC, 0.3 s.



05 · AGENT

Constraints and reasons

Elicits constraints from SOPs and operator speech, gated on human sign-off. Explains from the solver, not from itself.

~3,000 lines of Python, 45 regression tests passing, and every schedule re-simulated against the true nonlinear physics before anyone sees it.

The plant doesn't pay MCP - it pays the landed bill

Market price

+

Wheeling

+

Surcharges

+

Duty

+

Demand charge on peak kVA

=

Landed cost

The market clearing price is about 60% of what an industrial consumer pays. Wheeling, surcharges, duty and the demand charge on peak kVA are the other 46% of the bill so we put them inside the optimisation rather than after it.

SAME FIVE HIGH-SPREAD DAYS	OPTIMISED ON MCP	OPTIMISED ON LANDED COST	PENALTY	INDICATIVE PER YEAR
Chlor-alkali, 700 TPD	₹7.196 cr	₹7.128 cr	0.95%	₹5.0 cr
Refinery utilities block	₹9.167 cr	₹8.862 cr	3.33%	₹22.3 cr

Both schedules settled at the same real bill. Same physics, same days — only the price the optimiser was shown differs.

The regulator is pushing the same way. MERC's post-remand order removes the night rebate because it competes with shifting load into solar hours, and the demand charge rises 15% over four years while the energy charge falls 12%. Flexibility gets more valuable on the published trajectory, not less.

DEMONSTRATOR

Demonstrated on a 700 TPD chlor-alkali case

120 consecutive delivery days, replayed at settled prices, against the plant's real counterfactual: one steady setpoint per asset held all day.

₹0.133

net saving per kWh of total plant load, with real forecast error

₹9.17 cr

annualised ·
1.53% of the bill

0

violations · 0 MW
deviation used

13 / 120

days worse than
doing nothing

PRANA does not manufacture savings. When flexibility is not worthwhile it can lose: a single-asset site with a hard production floor has no second lever when the forecast is wrong. The 13 days are reported, not removed.

GROSS TO NET, ₹ LAKH OVER 120 DAYS

378.3



Gross benefit

-76.8



Process costs

301.5



Net benefit

Process cost is membrane wear, chlorine diverted to bleach, and kVAh droop one fifth of the gross, paid straight back. A model that does not charge these is not conservative, it is wrong.

120 delivery days to 2026-08-06, settled at realised prices. Perfect foresight would be ₹0.174/kWh; the forecast-error haircut is 22%. Figures are for this demonstrator plant, not a sector claim.

One demonstrator. A platform for industrial flexibility.

Chlor-alkali	DEMONSTRATED · 120 DAYS
Refinery	NEXT VALIDATION TARGET
Chemicals	VALIDATION TARGET
Fertilizer	VALIDATION TARGET
Hydrogen	DEPLOYMENT TARGET

No savings are claimed for any unvalidated industry. Each one earns its number the same way chlor-alkali did.

90-DAY ADVISORY PILOT

Weeks 1–4

Plant-specific digital twin, fitted to their historian

Weeks 5–8

Flexibility cost curve ϕ the number they have never had

Weeks 9–12

Shadow-mode optimisation, compared against actuals daily

Lower landed electricity cost

Zero product-spec excursions

Zero safety violations

No production authority transferred

PRANA does not tell the plant what equipment to sacrifice. It tells the plant what flexibility it can safely afford.